



Subject Vision and Intent: Computing

Our Vision

At King's Cross Academy, we believe that Computing gives children the knowledge, skills and confidence to understand, create and participate in an increasingly digital world. Our ambition is for children to leave KCA as curious, capable and responsible users of technology who understand something of how the digital world around them works and have the confidence to shape it rather than simply consume it.

Technology is an important part of children's lives, but familiarity with devices does not necessarily mean that children understand them. Our Computing curriculum therefore moves beyond simply using technology. We want children to understand how computers and systems work, how information can be created and communicated, and how computational thinking can be used to understand and solve problems.

Our Computing curriculum reflects our Academy vision, *Love Learning Together*, by encouraging children to explore, create, collaborate and solve problems. It develops our KCA values by giving children the courage to experiment, the endurance to debug and improve their work, and the integrity to make safe, responsible and ethical choices when using technology.

Our Intent

Our Computing curriculum is ambitious, inclusive and carefully sequenced. It identifies the essential knowledge, vocabulary, concepts and practical skills we want children to develop from the Early Years to Year 6, enabling them to become increasingly knowledgeable, independent and creative users of technology.

Our curriculum develops children's understanding across the interconnected areas of computer science, information technology and digital literacy.

Through computer science, children learn the principles that underpin computation and digital systems. They develop their understanding of algorithms, programming, logical reasoning, data and computer systems. Children learn to create and debug programs, identify patterns, break problems into smaller parts and develop increasingly efficient solutions.

Through information technology, children learn to use digital tools purposefully. They create, organise, store, manipulate and communicate digital content, selecting appropriate technology for different tasks. As they progress through the school, children become increasingly independent in choosing and combining digital tools to achieve a particular purpose.

Through digital literacy, children develop the knowledge and judgement needed to participate confidently, responsibly and safely in the digital world. They learn to consider the reliability of information, understand their digital footprint, protect their privacy and personal information, communicate appropriately and recognise when and how to seek help.

Online safety is woven throughout our Computing curriculum and wider curriculum rather than being taught as a single isolated topic. As technology develops and children's experiences change, we equip them with principles and strategies that help them make informed decisions, assess risk and behave responsibly online.

Computational thinking is developed progressively. Children learn to approach problems logically, identify patterns, create and follow sequences, test ideas and learn from mistakes. We want children to understand that when a program or solution does not work as intended, debugging, adapting and trying again are an important part of the process.

Vocabulary is explicitly taught and revisited so that children develop the language needed to explain digital systems, describe processes and communicate their thinking with increasing precision.

Every Child

We have high expectations for every child. Children with SEND, children who are disadvantaged and children who are developing English as an additional language access the same ambitious Computing curriculum.

Teachers identify the essential knowledge and skills that all children need and adapt teaching so that barriers are reduced without lowering expectations. Explicit modelling, carefully sequenced instructions, visual representations, appropriate technology, paired and collaborative work and opportunities for repeated practice enable children to develop confidence and increasing independence.

We recognise that children's access to technology and their experiences of the digital world outside school vary considerably. Our curriculum therefore ensures that every child has meaningful opportunities to use technology, develop computational thinking, create digital content and learn the knowledge they need to participate safely and confidently in the digital world.

We also recognise that technology will continue to change. Our curriculum therefore does not seek simply to teach children how to use particular devices or applications. Instead, we develop transferable knowledge, concepts and ways of thinking that children can apply as the technologies around them evolve.

By the end of Year 6, our aim is not simply that children can use technology, but that they increasingly understand it. We want them to create and debug programs, select and use digital tools purposefully, understand fundamental principles of computer systems and make informed and responsible choices in digital environments.

Above all, we want children to leave KCA as confident, creative and responsible digital citizens: children who are curious about how technology works, resilient when solving problems and equipped to use technology safely, critically and creatively in the world beyond KCA.